

Week 1: Project Architecture and Backend Setup

Project: RESQAI — AI-Assisted Humanitarian Disaster Message Classification

Project Architecture

The first week focused on establishing the overall backend architecture for the RESQAI disaster response application. The system was designed using a two-service architecture: a Node.js and Express backend for API handling, business logic, authentication, and database operations; and a Python FastAPI service for machine-learning inference and Gemini-based analysis. This separation keeps the ML components independent from the main application backend.

Backend Structure

The backend was organized into modules for configuration, controllers, middleware, models, routes, services, and utilities. Controllers were separated by responsibility: authentication, incident operations, administration, and rescue-team operations. Middleware was planned for JWT verification, role-based access control, and centralized error handling.

Database Design

MongoDB with Mongoose ODM was selected. Three primary collections were designed: Users, Incidents, and ClassificationResults. The Users collection stores authentication and role information, the Incidents collection stores disaster reports and their lifecycle, and ClassificationResults stores ML metadata for debugging and analysis.

Outcome

By the end of Week 1, the project architecture, backend structure, and database design were established, providing the foundation for authentication, incident processing, and ML integration.